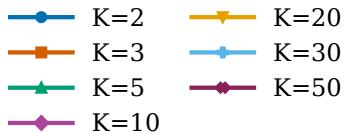
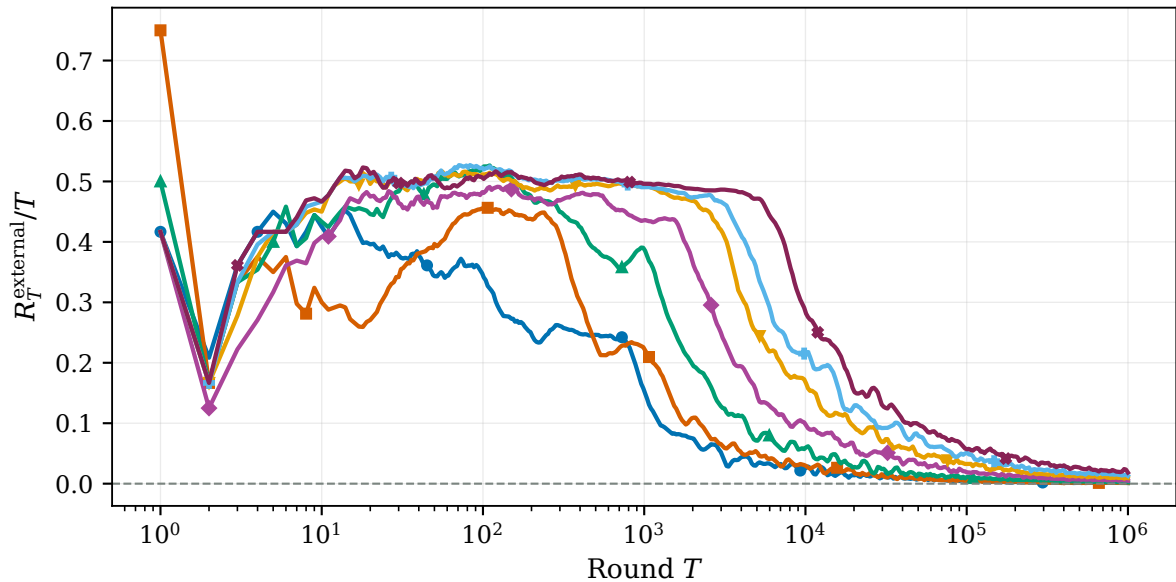


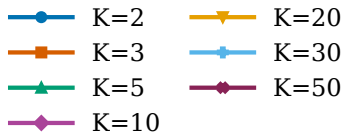
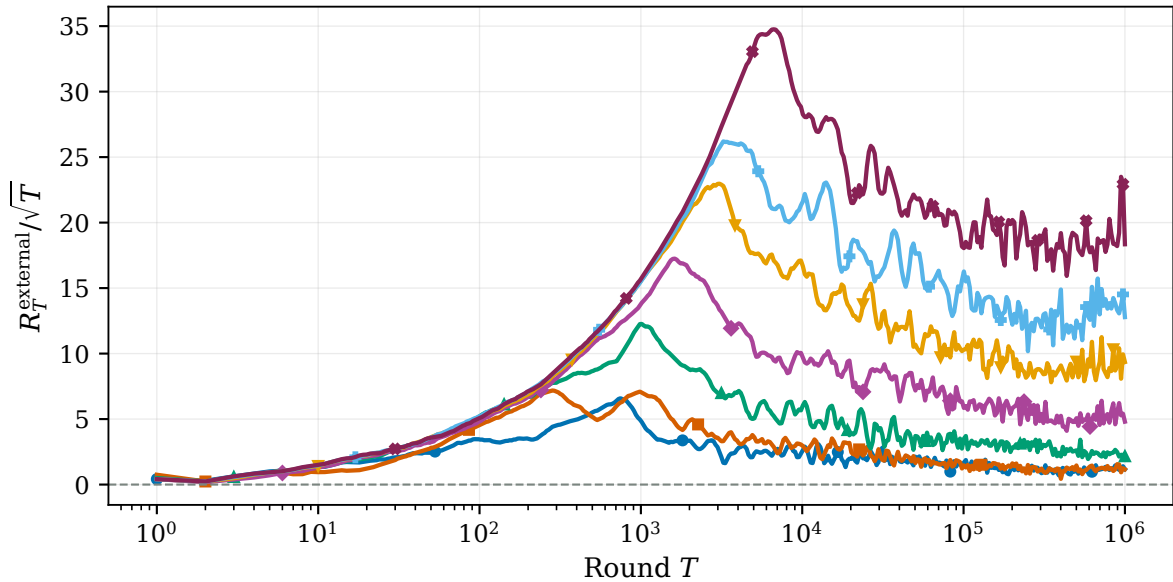
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: EXP3
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.00118067, SD = 0.000609149, SE = 0.000175846
K=3: mean = 0.00116583, SD = 0.00146873, SE = 0.000423985
K=5: mean = 0.002161, SD = 0.00101943, SE = 0.000294284
K=10: mean = 0.004803, SD = 0.00170114, SE = 0.000491078
K=20: mean = 0.009355, SD = 0.00205172, SE = 0.00059228
K=30: mean = 0.0127846, SD = 0.00264873, SE = 0.000764623
K=50: mean = 0.018346, SD = 0.00447744, SE = 0.00129252



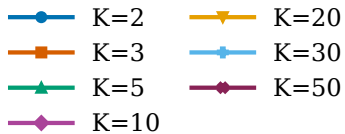
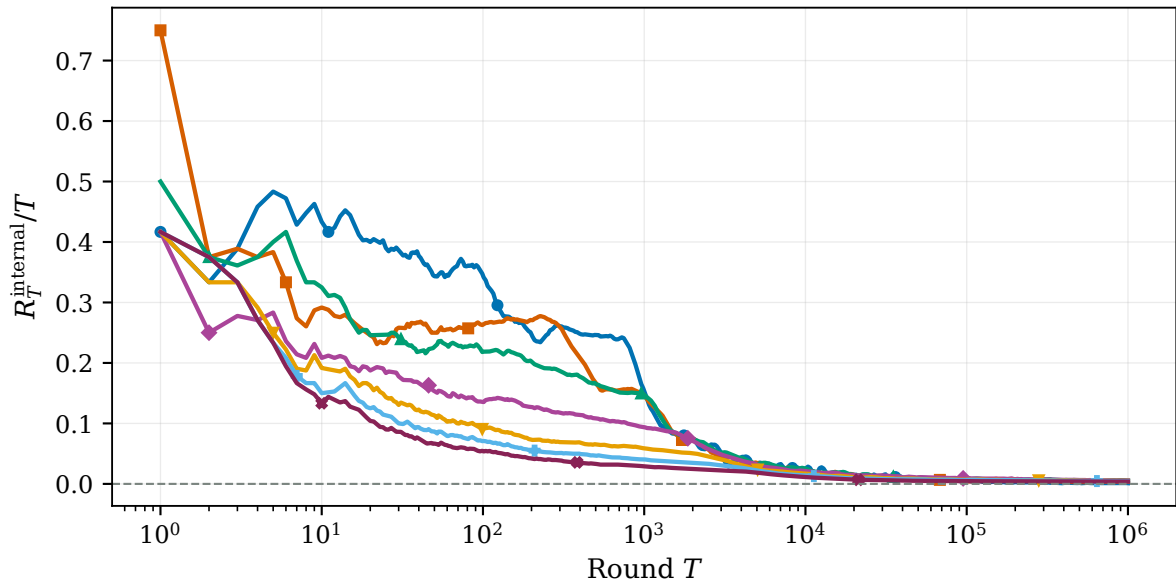
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: EXP3
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 1.18067, SD = 0.609149, SE = 0.175846
K=3: mean = 1.16583, SD = 1.46873, SE = 0.423985
K=5: mean = 2.161, SD = 1.01943, SE = 0.294284
K=10: mean = 4.803, SD = 1.70114, SE = 0.491078
K=20: mean = 9.355, SD = 2.05172, SE = 0.59228
K=30: mean = 12.7846, SD = 2.64873, SE = 0.764623
K=50: mean = 18.346, SD = 4.47744, SE = 1.29252



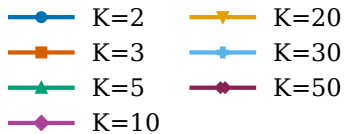
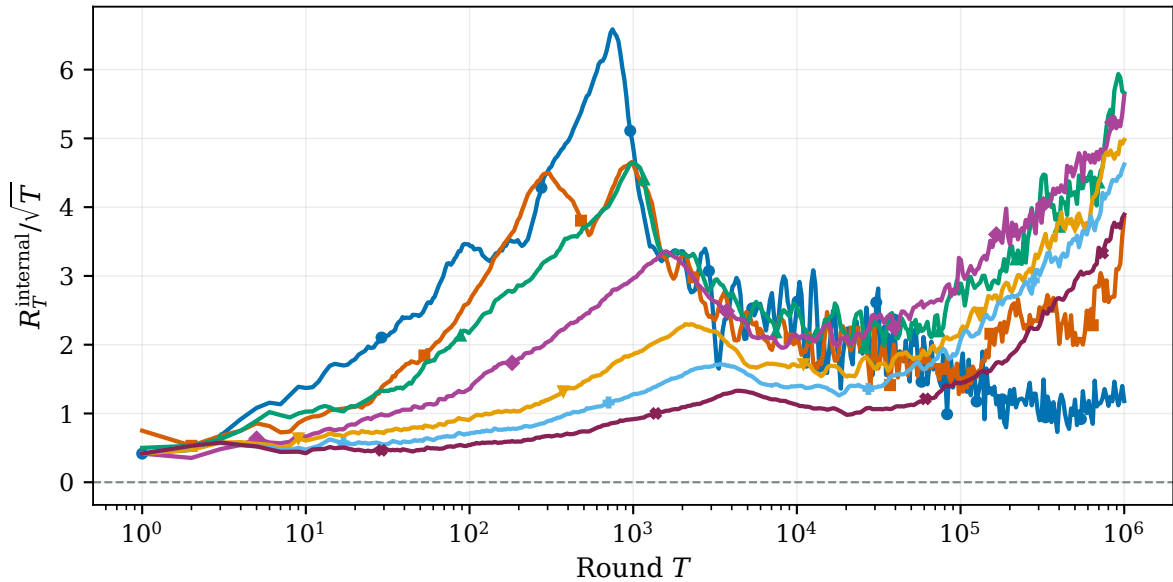
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: EXP3
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.00118067, SD = 0.000609149, SE = 0.000175846
K=3: mean = 0.00384517, SD = 0.00348931, SE = 0.00100728
K=5: mean = 0.00566017, SD = 0.00167124, SE = 0.000482445
K=10: mean = 0.00562192, SD = 0.00148798, SE = 0.000429542
K=20: mean = 0.0049765, SD = 0.000956117, SE = 0.000276007
K=30: mean = 0.00462042, SD = 0.000871075, SE = 0.000251458
K=50: mean = 0.00389133, SD = 0.000571033, SE = 0.000164843



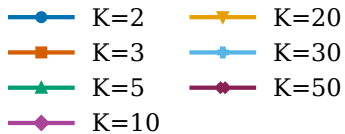
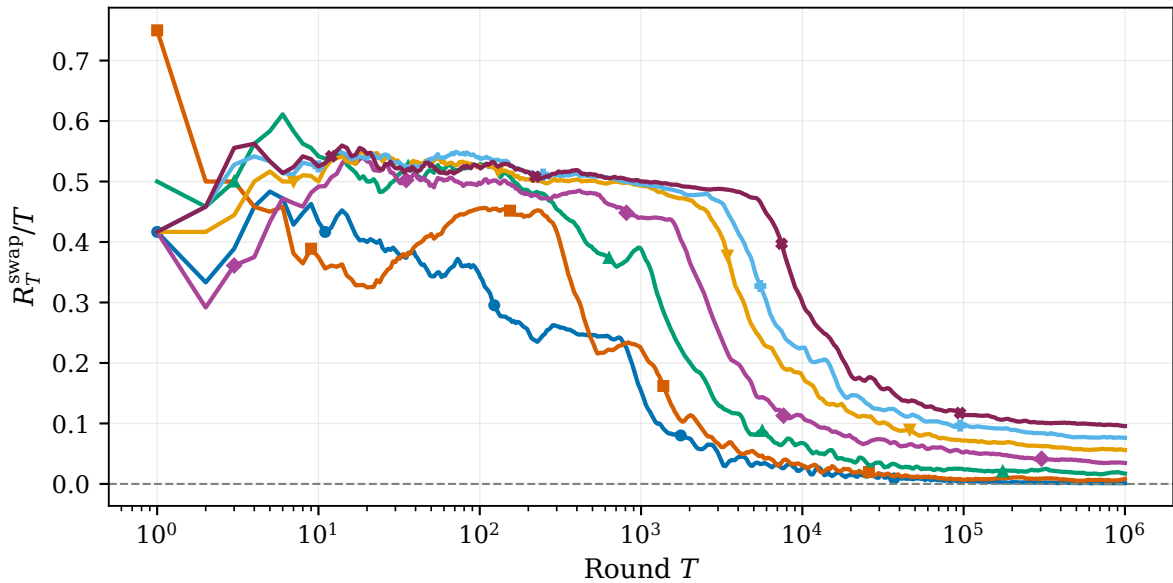
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: EXP3
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 1.18067, SD = 0.609149, SE = 0.175846
K=3: mean = 3.84517, SD = 3.48931, SE = 1.00728
K=5: mean = 5.66017, SD = 1.67124, SE = 0.482445
K=10: mean = 5.62192, SD = 1.48798, SE = 0.429542
K=20: mean = 4.9765, SD = 0.956117, SE = 0.276007
K=30: mean = 4.62042, SD = 0.871075, SE = 0.251458
K=50: mean = 3.89133, SD = 0.571033, SE = 0.164843



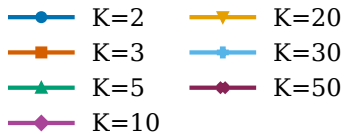
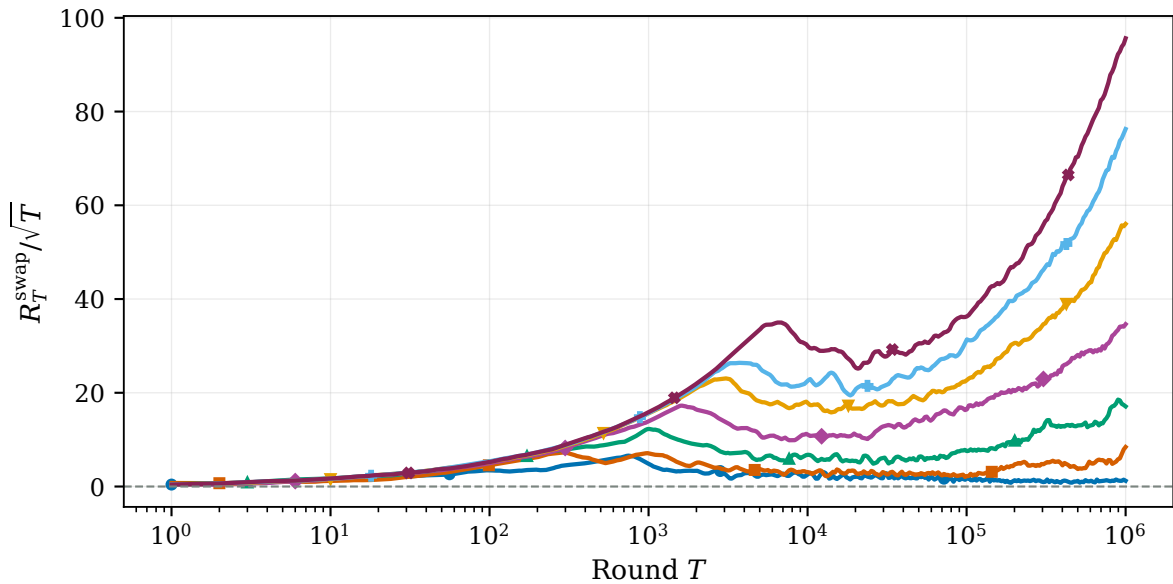
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: EXP3
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.00122233, SD = 0.000582708, SE = 0.000168213
K=3: mean = 0.00842783, SD = 0.00694864, SE = 0.0020059
K=5: mean = 0.0170904, SD = 0.00443012, SE = 0.00127886
K=10: mean = 0.0346296, SD = 0.00799226, SE = 0.00230717
K=20: mean = 0.0560176, SD = 0.00726672, SE = 0.00209772
K=30: mean = 0.0762643, SD = 0.00674811, SE = 0.00194801
K=50: mean = 0.0956253, SD = 0.00599973, SE = 0.00173197



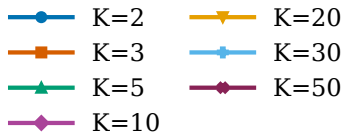
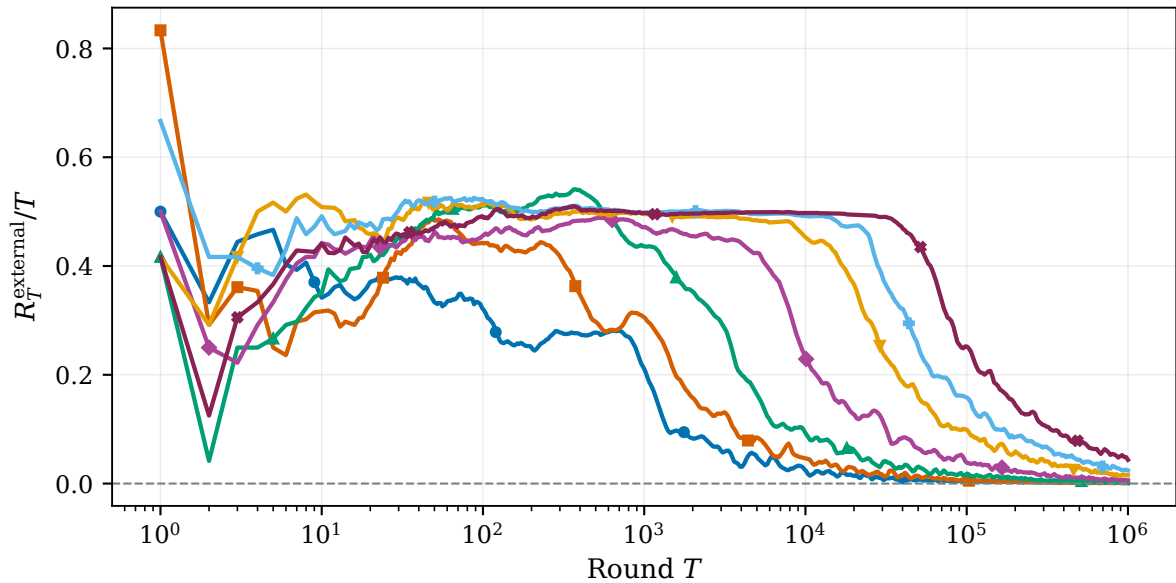
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: EXP3
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 1.22233, SD = 0.582708, SE = 0.168213
K=3: mean = 8.42783, SD = 6.94864, SE = 2.0059
K=5: mean = 17.0904, SD = 4.43012, SE = 1.27886
K=10: mean = 34.6296, SD = 7.99226, SE = 2.30717
K=20: mean = 56.0176, SD = 7.26672, SE = 2.09772
K=30: mean = 76.2643, SD = 6.74811, SE = 1.94801
K=50: mean = 95.6253, SD = 5.99973, SE = 1.73197



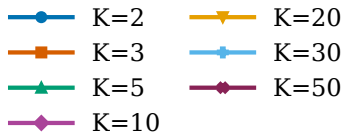
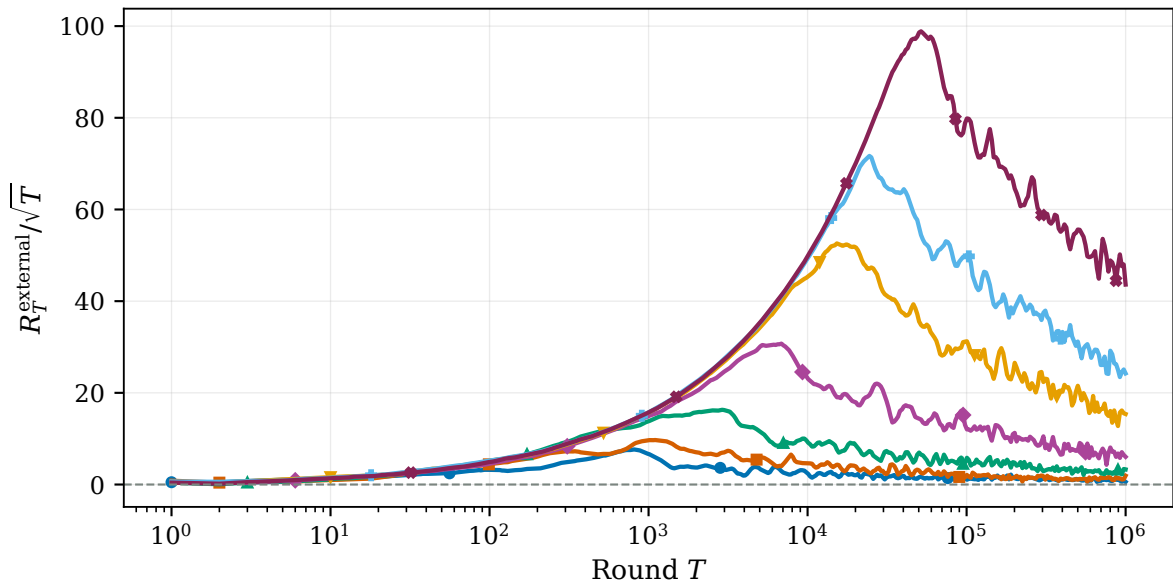
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: BM-EXP3
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000572083, SD = 0.000688454, SE = 0.00019874
K=3: mean = 0.00192775, SD = 0.00122494, SE = 0.000353608
K=5: mean = 0.0032975, SD = 0.00111353, SE = 0.00032145
K=10: mean = 0.00603992, SD = 0.00179243, SE = 0.000517429
K=20: mean = 0.0153785, SD = 0.00260661, SE = 0.000752462
K=30: mean = 0.0242372, SD = 0.00561265, SE = 0.00162023
K=50: mean = 0.0436196, SD = 0.0104343, SE = 0.00301212



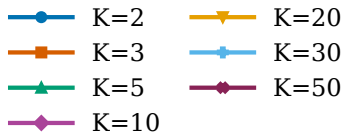
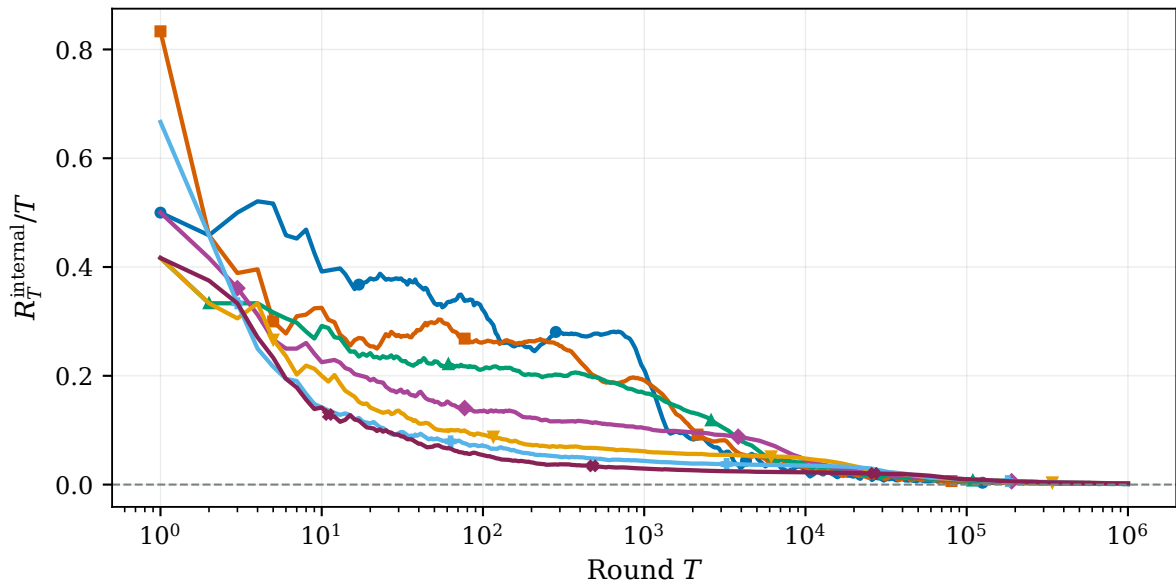
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: BM-EXP3
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.572083, SD = 0.688454, SE = 0.19874
K=3: mean = 1.92775, SD = 1.22494, SE = 0.353608
K=5: mean = 3.2975, SD = 1.11353, SE = 0.32145
K=10: mean = 6.03992, SD = 1.79243, SE = 0.517429
K=20: mean = 15.3785, SD = 2.60661, SE = 0.752462
K=30: mean = 24.2372, SD = 5.61265, SE = 1.62023
K=50: mean = 43.6196, SD = 10.4343, SE = 3.01212



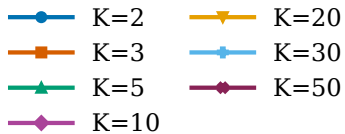
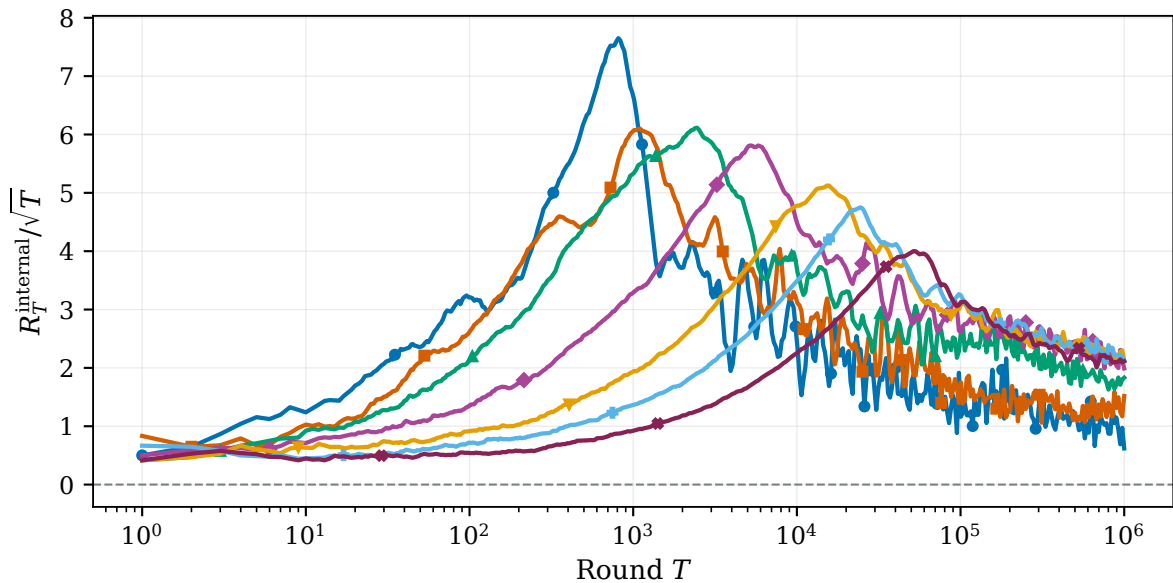
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: BM-EXP3
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000619333, SD = 0.000619137, SE = 0.000178729
K=3: mean = 0.00151867, SD = 0.00087411, SE = 0.000252334
K=5: mean = 0.00183267, SD = 0.000500937, SE = 0.000144608
K=10: mean = 0.00199142, SD = 0.000436228, SE = 0.000125928
K=20: mean = 0.00212042, SD = 0.000321253, SE = 9.27377e-05
K=30: mean = 0.00212067, SD = 0.00020892, SE = 6.03101e-05
K=50: mean = 0.00211283, SD = 0.000212359, SE = 6.13028e-05



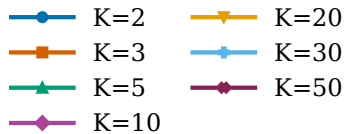
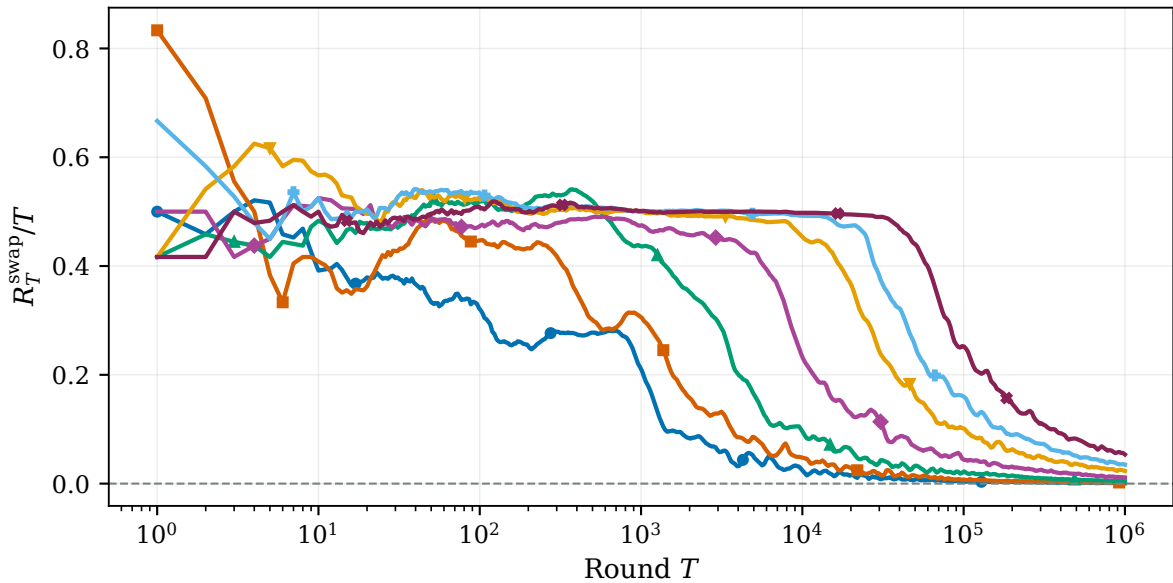
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: BM-EXP3
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.619333, SD = 0.619137, SE = 0.178729
K=3: mean = 1.51867, SD = 0.87411, SE = 0.252334
K=5: mean = 1.83267, SD = 0.500937, SE = 0.144608
K=10: mean = 1.99142, SD = 0.436228, SE = 0.125928
K=20: mean = 2.12042, SD = 0.321253, SE = 0.0927377
K=30: mean = 2.12067, SD = 0.20892, SE = 0.0603101
K=50: mean = 2.11283, SD = 0.212359, SE = 0.0613028



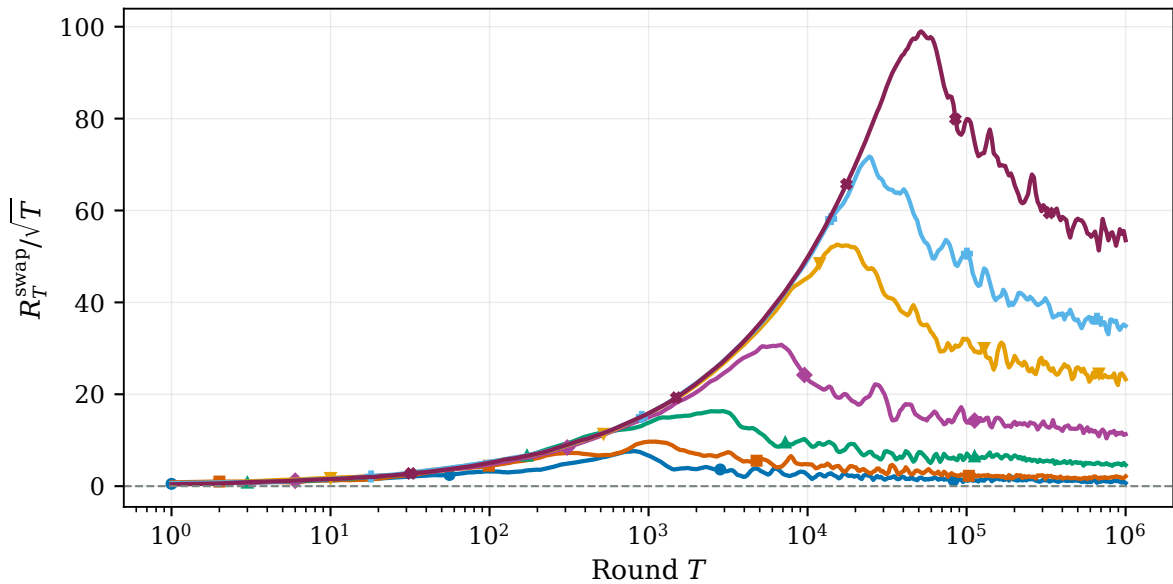
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: BM-EXP3
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.00066475, SD = 0.000637486, SE = 0.000184026
K=3: mean = 0.00214442, SD = 0.00104008, SE = 0.000300247
K=5: mean = 0.00461583, SD = 0.000963445, SE = 0.000278123
K=10: mean = 0.0113095, SD = 0.00147638, SE = 0.000426195
K=20: mean = 0.0232643, SD = 0.00122476, SE = 0.000353559
K=30: mean = 0.0348854, SD = 0.00399876, SE = 0.00115434
K=50: mean = 0.0535373, SD = 0.00690531, SE = 0.00199339



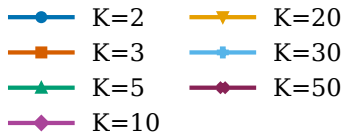
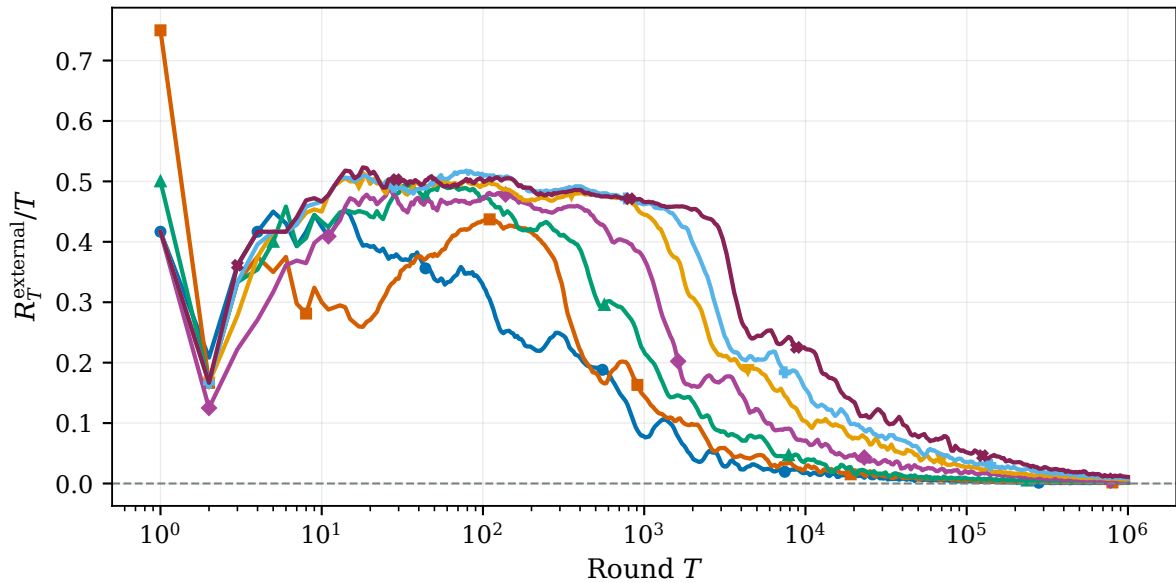
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: BM-EXP3
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.66475, SD = 0.637486, SE = 0.184026
K=3: mean = 2.14442, SD = 1.04008, SE = 0.300247
K=5: mean = 4.61583, SD = 0.963445, SE = 0.278123
K=10: mean = 11.3095, SD = 1.47638, SE = 0.426195
K=20: mean = 23.2643, SD = 1.22476, SE = 0.353559
K=30: mean = 34.8854, SD = 3.99876, SE = 1.15434
K=50: mean = 53.5373, SD = 6.90531, SE = 1.99339



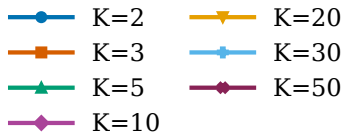
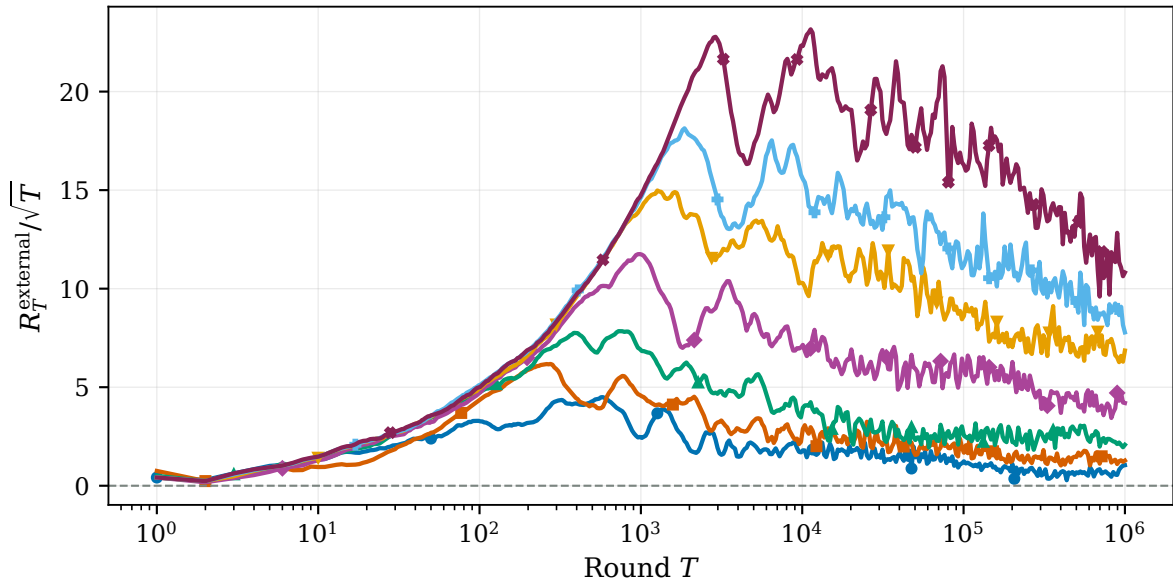
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: EXP3-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.00104275, SD = 0.000785767, SE = 0.000226831
K=3: mean = 0.001278, SD = 0.00106971, SE = 0.000308798
K=5: mean = 0.00207817, SD = 0.000940928, SE = 0.000271622
K=10: mean = 0.00419725, SD = 0.0025064, SE = 0.000723535
K=20: mean = 0.00685892, SD = 0.0015171, SE = 0.000437949
K=30: mean = 0.0077825, SD = 0.00191758, SE = 0.000553558
K=50: mean = 0.0108213, SD = 0.00268336, SE = 0.000774621



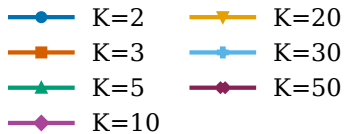
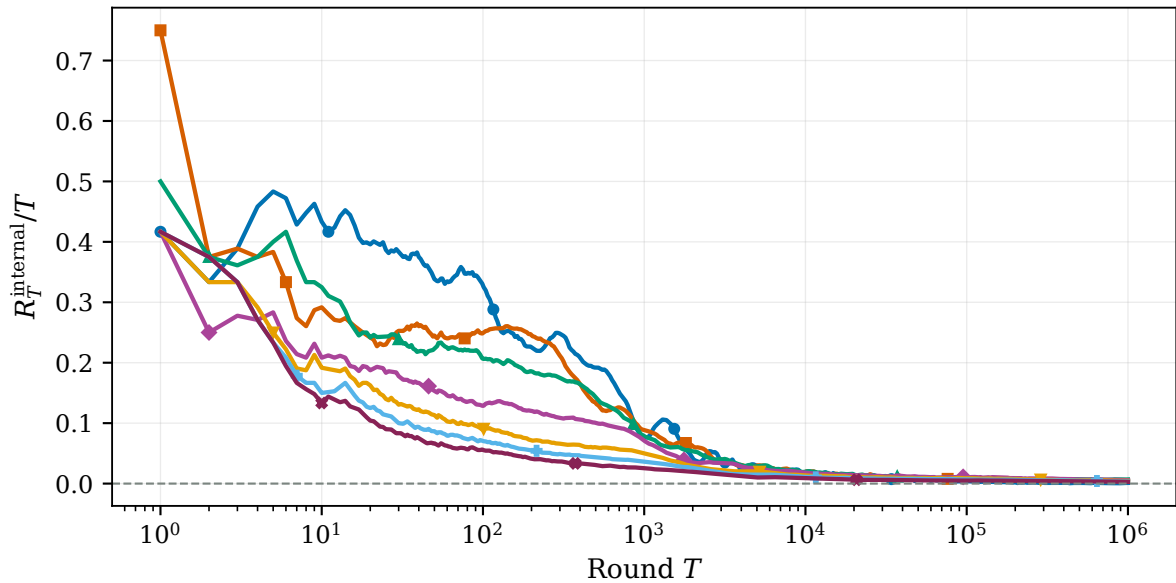
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: EXP3-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 1.04275, SD = 0.785767, SE = 0.226831
K=3: mean = 1.278, SD = 1.06971, SE = 0.308798
K=5: mean = 2.07817, SD = 0.940928, SE = 0.271622
K=10: mean = 4.19725, SD = 2.5064, SE = 0.723535
K=20: mean = 6.85892, SD = 1.5171, SE = 0.437949
K=30: mean = 7.7825, SD = 1.91758, SE = 0.553558
K=50: mean = 10.8213, SD = 2.68336, SE = 0.774621



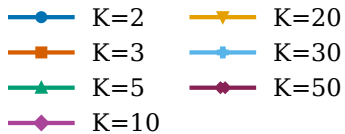
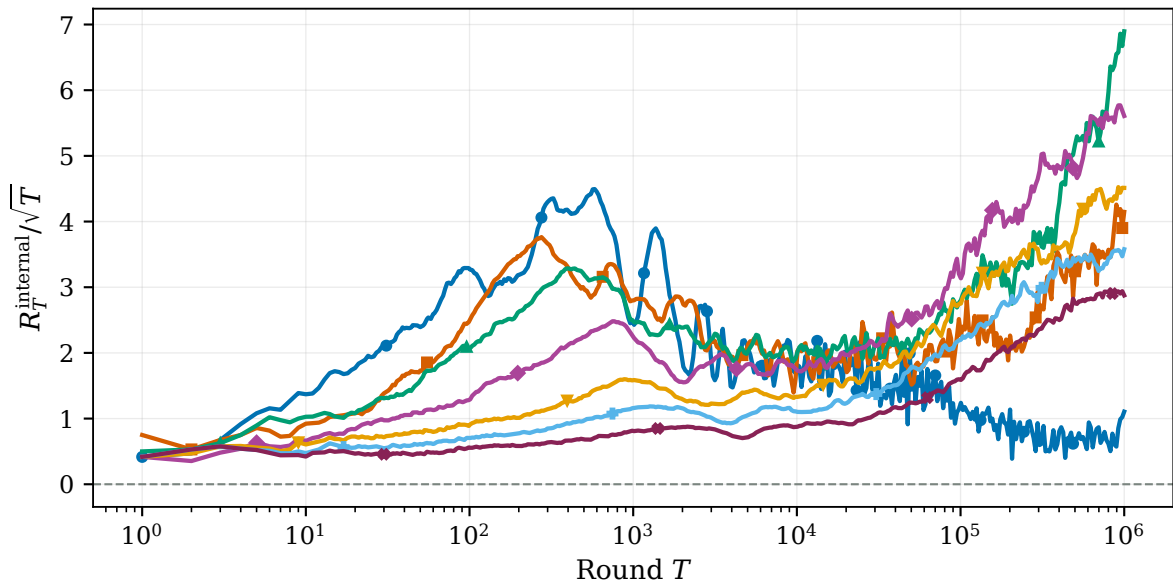
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: EXP3-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.00110542, SD = 0.000647429, SE = 0.000186897
K=3: mean = 0.0041465, SD = 0.00180603, SE = 0.000521356
K=5: mean = 0.00689642, SD = 0.000803421, SE = 0.000231928
K=10: mean = 0.00560775, SD = 0.00108382, SE = 0.000312871
K=20: mean = 0.00451283, SD = 0.000573019, SE = 0.000165416
K=30: mean = 0.00357825, SD = 0.000667204, SE = 0.000192605
K=50: mean = 0.0028755, SD = 0.000361909, SE = 0.000104474



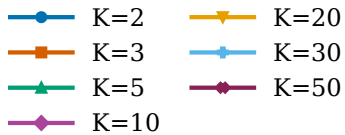
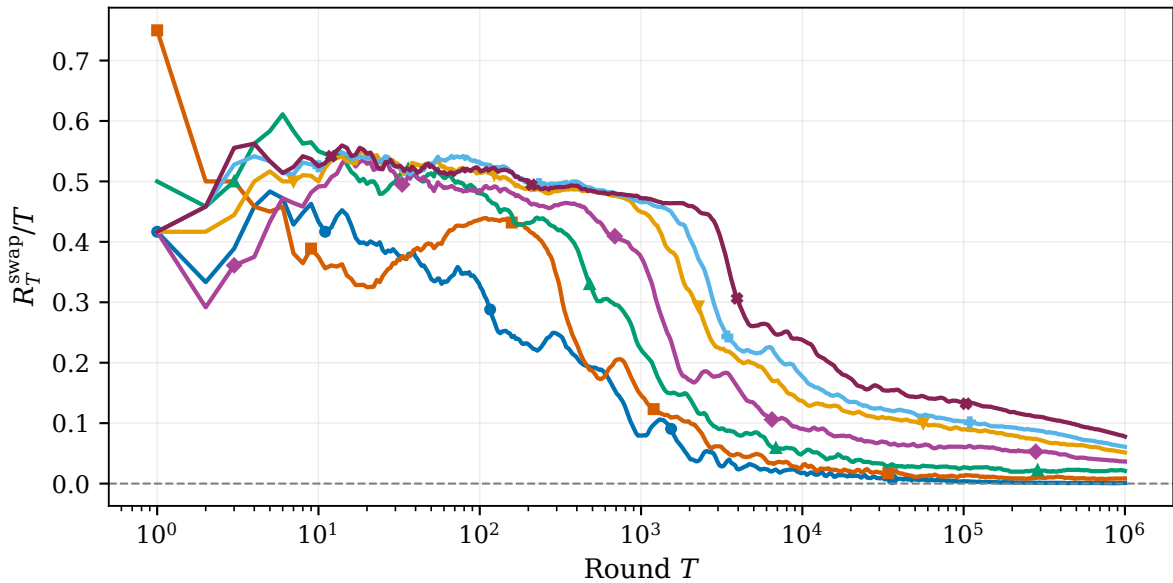
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: EXP3-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 1.10542, SD = 0.647429, SE = 0.186897
K=3: mean = 4.1465, SD = 1.80603, SE = 0.521356
K=5: mean = 6.89642, SD = 0.803421, SE = 0.231928
K=10: mean = 5.60775, SD = 1.08382, SE = 0.312871
K=20: mean = 4.51283, SD = 0.573019, SE = 0.165416
K=30: mean = 3.57825, SD = 0.667204, SE = 0.192605
K=50: mean = 2.8755, SD = 0.361909, SE = 0.104474



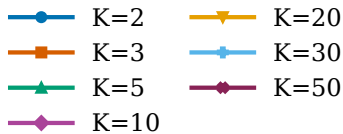
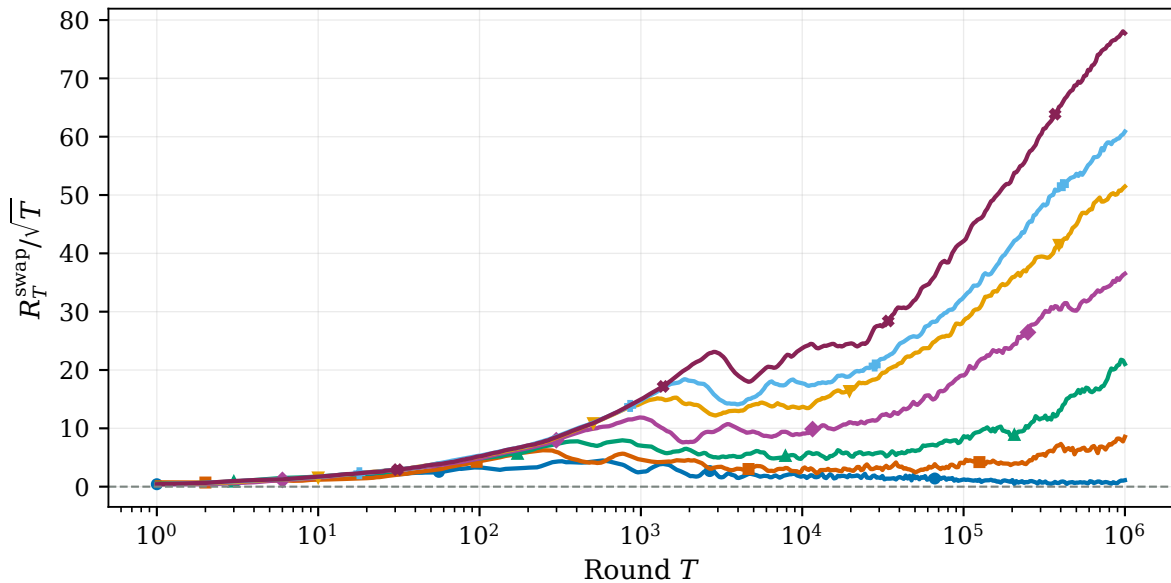
Experiment Information

Figure:	Action-space comparison
Environment:	Historical-frequency
Actions:	2, 3, 5, 10, 20, 30, 50
Feedback:	Bandit feedback
Algorithm:	EXP3-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2:	mean = 0.00110542, SD = 0.000647429, SE = 0.000186897
K=3:	mean = 0.00854808, SD = 0.00357063, SE = 0.00103075
K=5:	mean = 0.0210296, SD = 0.00325854, SE = 0.00094066
K=10:	mean = 0.036513, SD = 0.00761423, SE = 0.00219804
K=20:	mean = 0.0514625, SD = 0.00382154, SE = 0.00110318
K=30:	mean = 0.0608813, SD = 0.00489516, SE = 0.00141311
K=50:	mean = 0.077744, SD = 0.00439211, SE = 0.00126789



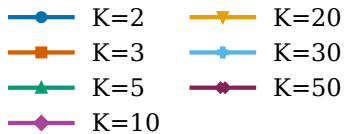
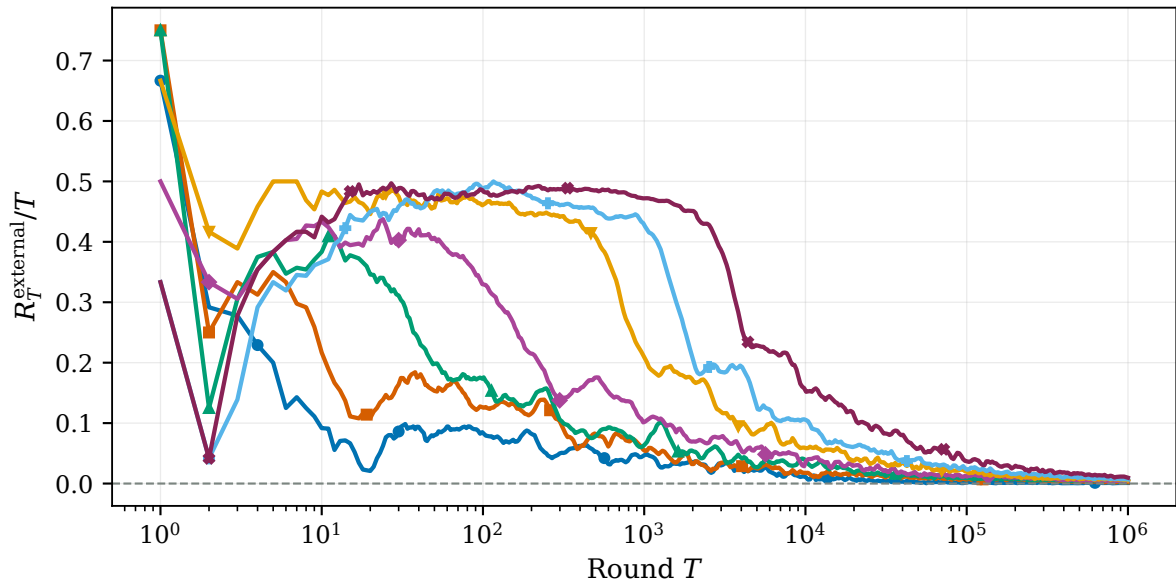
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: EXP3-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 1.10542, SD = 0.647429, SE = 0.186897
K=3: mean = 8.54808, SD = 3.57063, SE = 1.03075
K=5: mean = 21.0296, SD = 3.25854, SE = 0.94066
K=10: mean = 36.513, SD = 7.61423, SE = 2.19804
K=20: mean = 51.4625, SD = 3.82154, SE = 1.10318
K=30: mean = 60.8813, SD = 4.89516, SE = 1.41311
K=50: mean = 77.744, SD = 4.39211, SE = 1.26789



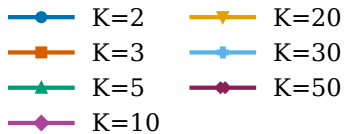
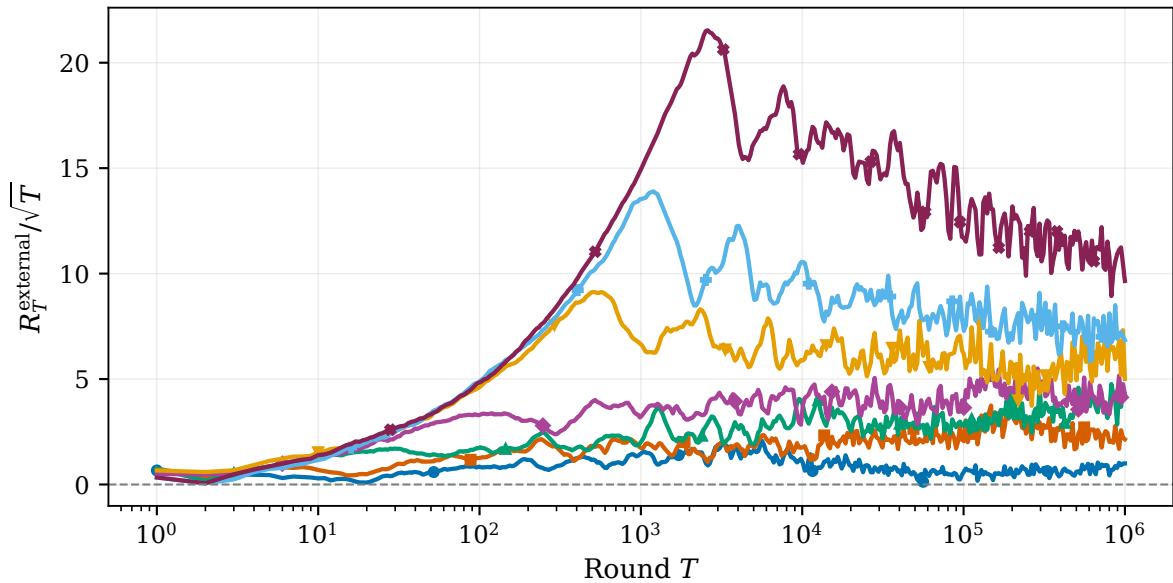
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: Ito-Tsallis
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000998417, SD = 0.00080684, SE = 0.000232915
K=3: mean = 0.00214217, SD = 0.00136867, SE = 0.000395101
K=5: mean = 0.00422592, SD = 0.0032321, SE = 0.000933027
K=10: mean = 0.0042845, SD = 0.00121424, SE = 0.000350521
K=20: mean = 0.00500842, SD = 0.00268013, SE = 0.000773687
K=30: mean = 0.006837, SD = 0.00305214, SE = 0.000881078
K=50: mean = 0.00965092, SD = 0.00280064, SE = 0.000808476



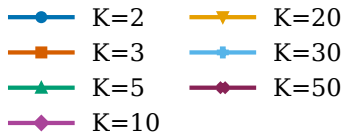
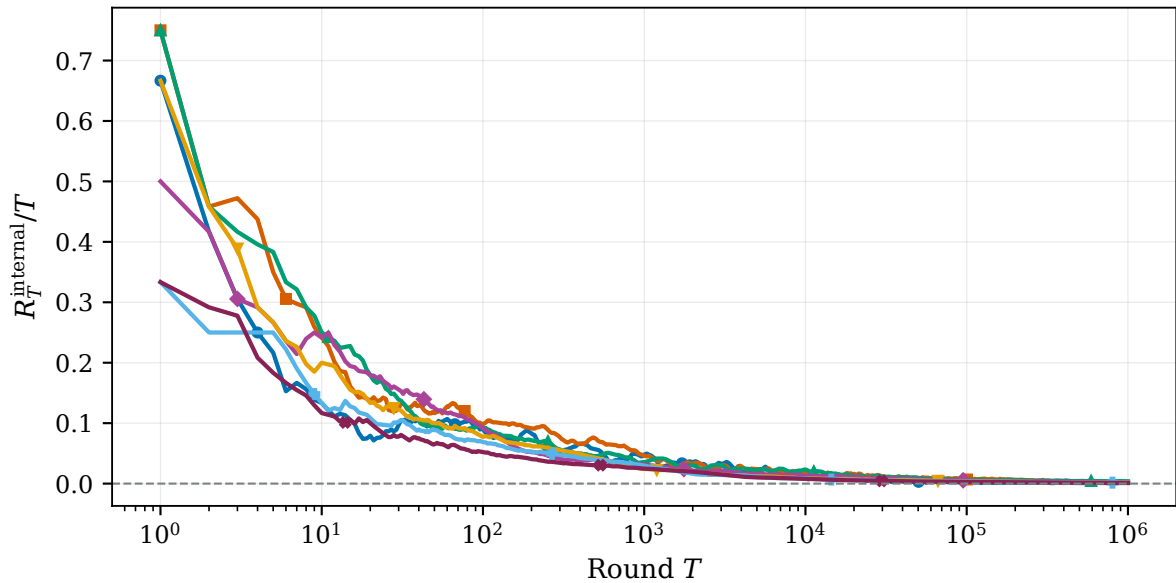
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: Ito-Tsallis
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.998417, SD = 0.80684, SE = 0.232915
K=3: mean = 2.14217, SD = 1.36867, SE = 0.395101
K=5: mean = 4.22592, SD = 3.2321, SE = 0.933027
K=10: mean = 4.2845, SD = 1.21424, SE = 0.350521
K=20: mean = 5.00842, SD = 2.68013, SE = 0.773687
K=30: mean = 6.837, SD = 3.05214, SE = 0.881078
K=50: mean = 9.65092, SD = 2.80064, SE = 0.808476



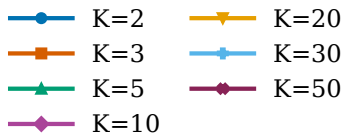
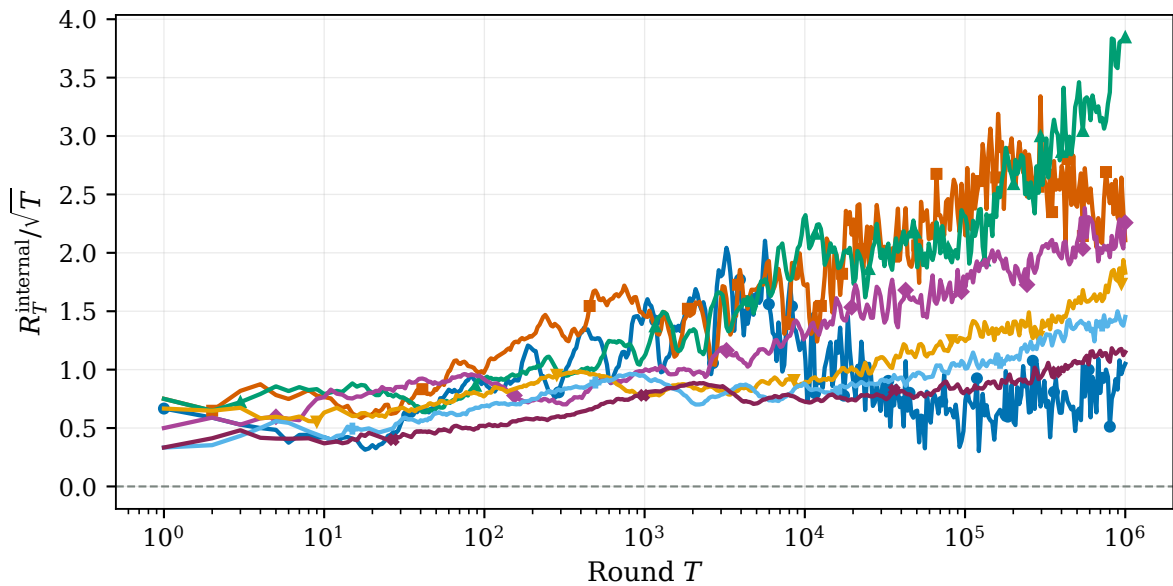
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: Ito-Tsallis
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.00105, SD = 0.000707734, SE = 0.000204305
K=3: mean = 0.00210808, SD = 0.00101409, SE = 0.000292744
K=5: mean = 0.00384433, SD = 0.00433555, SE = 0.00125157
K=10: mean = 0.00225817, SD = 0.000532163, SE = 0.000153622
K=20: mean = 0.0018305, SD = 0.000278471, SE = 8.03877e-05
K=30: mean = 0.00145183, SD = 0.000147382, SE = 4.25455e-05
K=50: mean = 0.00115058, SD = 0.000171699, SE = 4.95653e-05



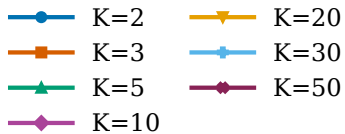
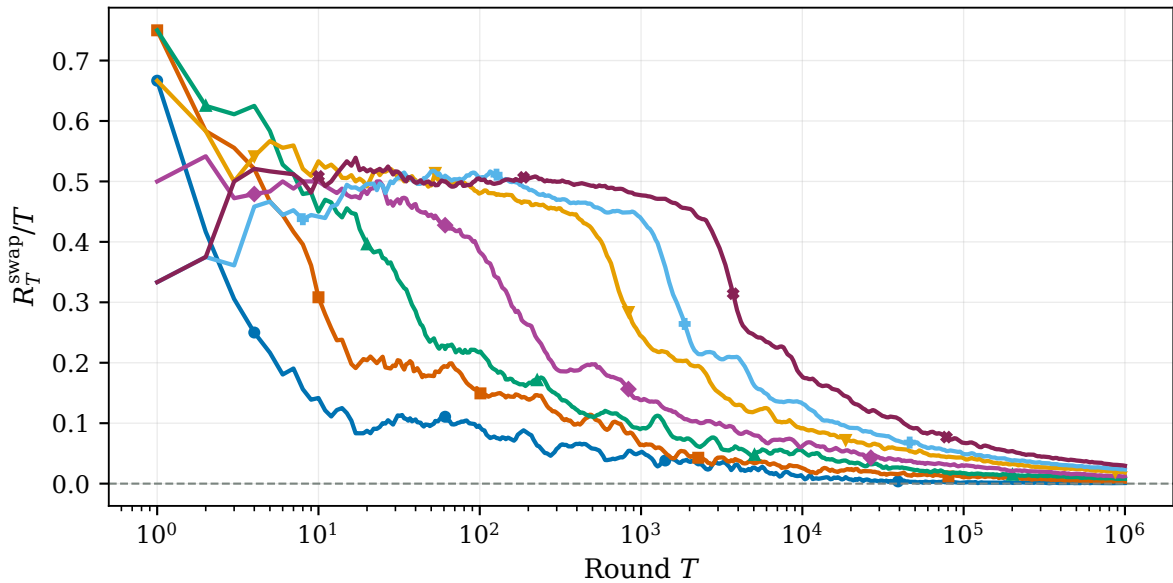
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: Ito-Tsallis
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 1.05, SD = 0.707734, SE = 0.204305
K=3: mean = 2.10808, SD = 1.01409, SE = 0.292744
K=5: mean = 3.84433, SD = 4.33555, SE = 1.25157
K=10: mean = 2.25817, SD = 0.532163, SE = 0.153622
K=20: mean = 1.8305, SD = 0.278471, SE = 0.0803877
K=30: mean = 1.45183, SD = 0.147382, SE = 0.0425455
K=50: mean = 1.15058, SD = 0.171699, SE = 0.0495653



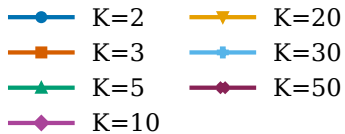
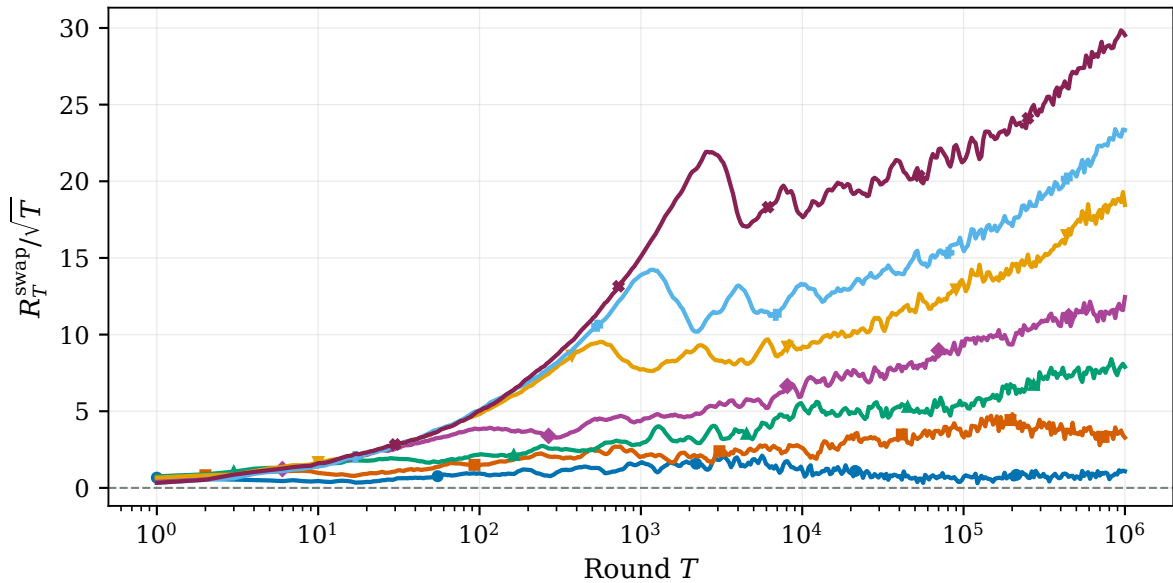
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: Ito-Tsallis
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.00108633, SD = 0.000741956, SE = 0.000214184
K=3: mean = 0.00327925, SD = 0.00182496, SE = 0.00052682
K=5: mean = 0.00789833, SD = 0.00516, SE = 0.00148956
K=10: mean = 0.0124485, SD = 0.00251438, SE = 0.000725839
K=20: mean = 0.0184529, SD = 0.00174114, SE = 0.000502623
K=30: mean = 0.0233418, SD = 0.00147447, SE = 0.000425643
K=50: mean = 0.0295365, SD = 0.00154344, SE = 0.000445553



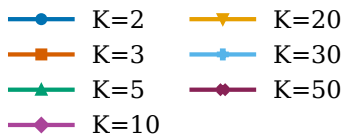
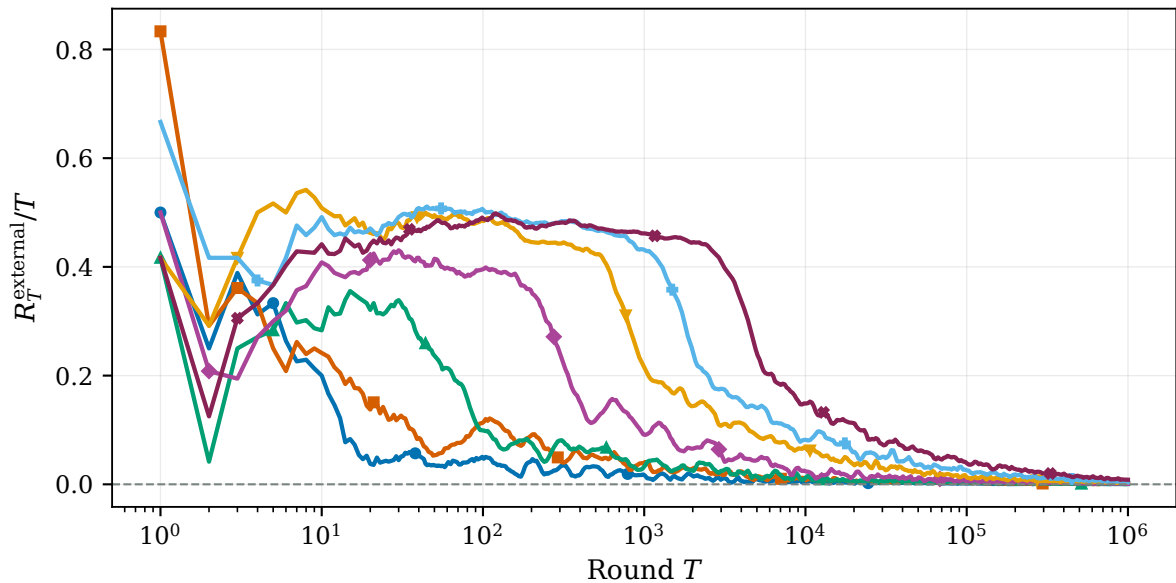
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: Ito-Tsallis
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 1.08633, SD = 0.741956, SE = 0.214184
K=3: mean = 3.27925, SD = 1.82496, SE = 0.52682
K=5: mean = 7.89833, SD = 5.16, SE = 1.48956
K=10: mean = 12.4485, SD = 2.51438, SE = 0.725839
K=20: mean = 18.4529, SD = 1.74114, SE = 0.502623
K=30: mean = 23.3418, SD = 1.47447, SE = 0.425643
K=50: mean = 29.5365, SD = 1.54344, SE = 0.445553



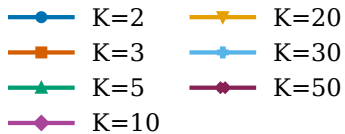
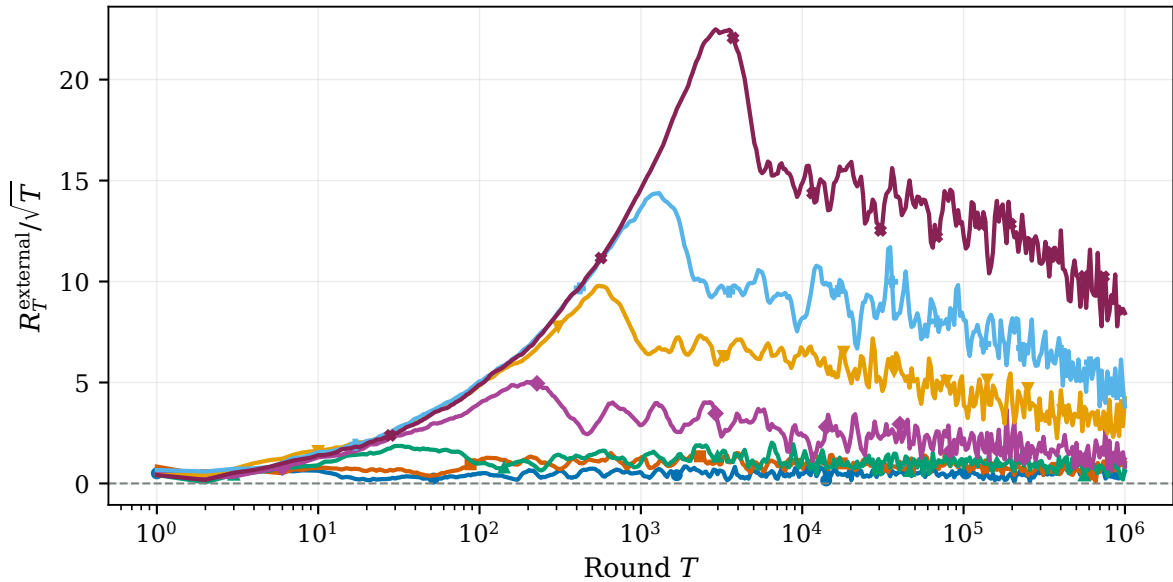
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: LCE-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000592333, SD = 0.000578747, SE = 0.00016707
K=3: mean = 0.001049, SD = 0.000980229, SE = 0.000282968
K=5: mean = 0.0009115, SD = 0.00123292, SE = 0.000355912
K=10: mean = 0.000834583, SD = 0.00119436, SE = 0.000344781
K=20: mean = 0.00421425, SD = 0.00214031, SE = 0.000617856
K=30: mean = 0.00382042, SD = 0.00198478, SE = 0.000572956
K=50: mean = 0.00843567, SD = 0.00386874, SE = 0.00111681



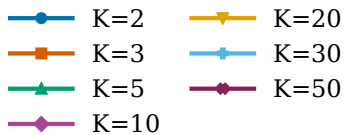
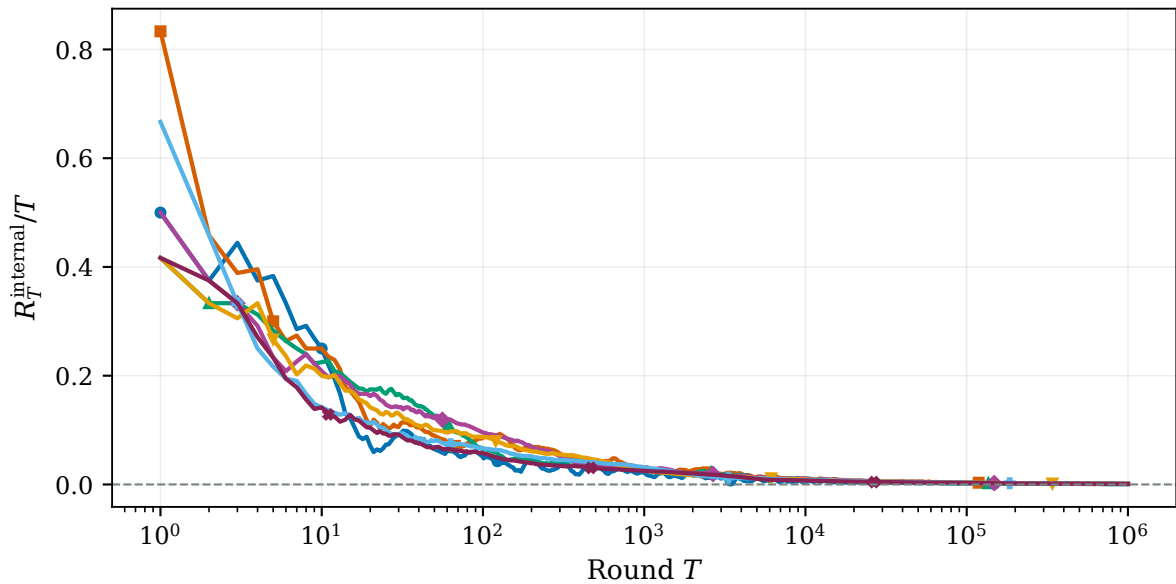
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: LCE-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.592333, SD = 0.578747, SE = 0.16707
K=3: mean = 1.049, SD = 0.980229, SE = 0.282968
K=5: mean = 0.9115, SD = 1.23292, SE = 0.355912
K=10: mean = 0.834583, SD = 1.19436, SE = 0.344781
K=20: mean = 4.21425, SD = 2.14031, SE = 0.617856
K=30: mean = 3.82042, SD = 1.98478, SE = 0.572956
K=50: mean = 8.43567, SD = 3.86874, SE = 1.11681



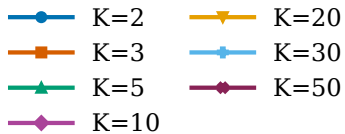
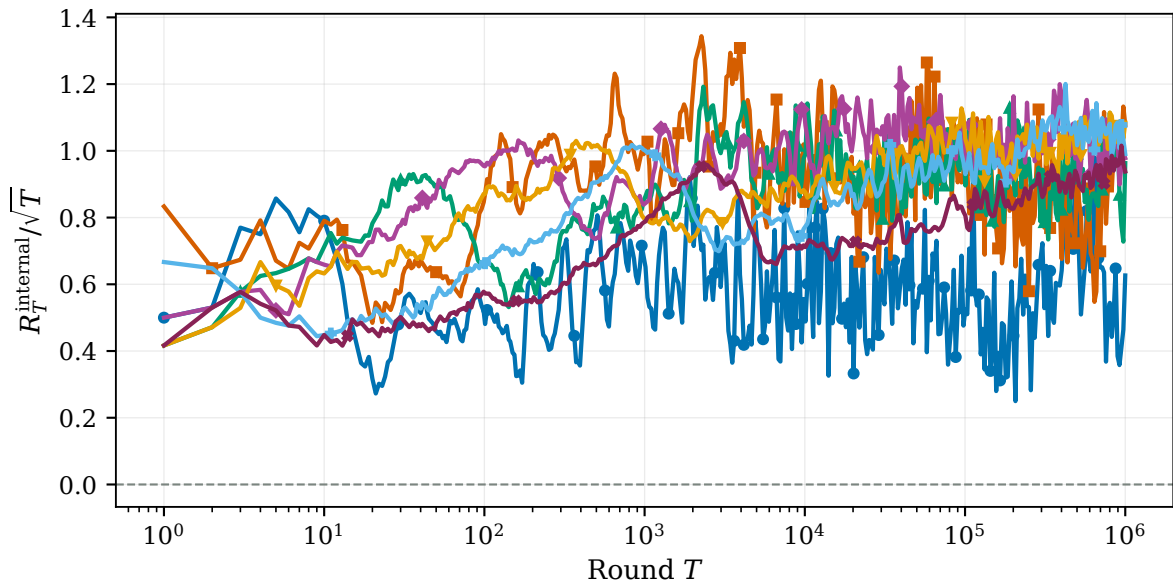
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: LCE-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000626417, SD = 0.000523856, SE = 0.000151224
K=3: mean = 0.0010775, SD = 0.000542026, SE = 0.00015647
K=5: mean = 0.000963083, SD = 0.000373782, SE = 0.000107901
K=10: mean = 0.000978167, SD = 0.000230327, SE = 6.64896e-05
K=20: mean = 0.00108467, SD = 0.000118277, SE = 3.41437e-05
K=30: mean = 0.000990417, SD = 0.00010931, SE = 3.1555e-05
K=50: mean = 0.000938083, SD = 0.000110447, SE = 3.18834e-05



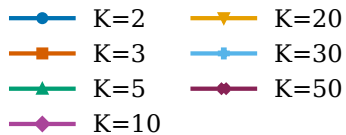
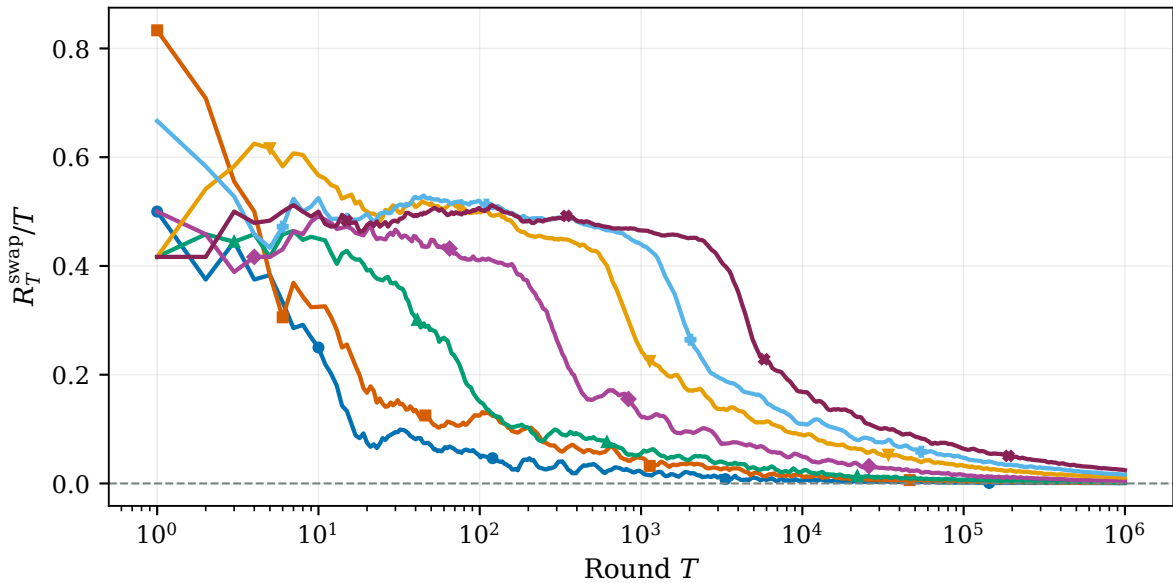
Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: LCE-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.626417, SD = 0.523856, SE = 0.151224
K=3: mean = 1.0775, SD = 0.542026, SE = 0.15647
K=5: mean = 0.963083, SD = 0.373782, SE = 0.107901
K=10: mean = 0.978167, SD = 0.230327, SE = 0.0664896
K=20: mean = 1.08467, SD = 0.118277, SE = 0.0341437
K=30: mean = 0.990417, SD = 0.10931, SE = 0.031555
K=50: mean = 0.938083, SD = 0.110447, SE = 0.0318834



Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: LCE-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.000649333, SD = 0.000544111, SE = 0.000157071
K=3: mean = 0.00140808, SD = 0.00080452, SE = 0.000232245
K=5: mean = 0.00190333, SD = 0.00108179, SE = 0.000312287
K=10: mean = 0.00415367, SD = 0.00111991, SE = 0.000323289
K=20: mean = 0.0109655, SD = 0.00111604, SE = 0.000322173
K=30: mean = 0.0162822, SD = 0.00128978, SE = 0.000372328
K=50: mean = 0.0243708, SD = 0.00163029, SE = 0.000470623



Experiment Information

Figure: Action-space comparison
Environment: Historical-frequency
Actions: 2, 3, 5, 10, 20, 30, 50
Feedback: Bandit feedback
Algorithm: LCE-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: $R / \sqrt{T}$
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
K=2: mean = 0.649333, SD = 0.544111, SE = 0.157071
K=3: mean = 1.40808, SD = 0.80452, SE = 0.232245
K=5: mean = 1.90333, SD = 1.08179, SE = 0.312287
K=10: mean = 4.15367, SD = 1.11991, SE = 0.323289
K=20: mean = 10.9655, SD = 1.11604, SE = 0.322173
K=30: mean = 16.2822, SD = 1.28978, SE = 0.372328
K=50: mean = 24.3708, SD = 1.63029, SE = 0.470623

